

SPEC - D1: where the hole is - the map's green centre, corrected by his pin sheet

Fable, 2026-09-28, at xhigh (Matt: "build D1"; `get_session self read xhigh` before a line of this was written). An xhigh class: the strokes-gained engine's input and the data model. Two builds from this one file:

Part	What	Who builds	Build
A	The engine's hole position, the data model key, the derived numbers, what the screens say about the source	Opus at xhigh (<code>.claude/agents/opus-xhigh.md</code>)	v33
B	The PIN SHEET sheet he types the numbers into	Opus at high (<code>.claude/agents/opus.md</code>), after Fable has signed Part A	v34

Fable reviews each diff against this file at xhigh and signs in `docs/DECISIONS_LOG.md`.

His words that bound it, verbatim:

- 2026-09-28, the ruling: *"D1. map center with the option for me to correct it manually by entering tournament pin sheet numbers. Build in whatever order I dont fucking care you ask so many questions"*
- 2026-09-28, his item 3, answered "yes": distance to the hole uses the map's green, never a cup taken from the track.
- 2026-09-28, his item 1, answered *"yes that is what I meant"*: shot 1 is the tee box at the card yardage. His comment on hole 14: *"needs to default to the scorecard"*.
- 2026-09-26, quoted in `js/ui/screen-play.js` `shotRowHeading`: *"center of the green and I can enter pin sheet distances in manually later"* and *"Numbers are always measured with distance to hole not the last shot or any other garbage."*
- After field test 4, quoted in `js/round/track-analysis.js`: *"enter a rough number of paces the cup was located on the green like a tournament pin sheet does."*
- 2026-09-28: *"I am not worried about yesterdays round."* No stored round is rewritten by this job.

Provenance tags: `[measured]` read from code or data this session, `[design]` this spec's decision, `[verify]` Opus confirms during the build.

One reading this spec makes, stated so he can correct it: a tournament pin sheet gives two numbers - paces ON from the front edge, and paces from the nearer SIDE EDGE, left or right. The entry stores which convention it was typed in (`sideFrom`), so if his sheets read from the middle instead, the change is one constant and no stored number is reinterpreted.

0. What this is, and what it is not

Is: the answer to "where is the hole" for every distance the app shows and every distance the strokes-gained engine looks up, on a course that has a map. No marked cup: the centre of the map's green. His pin sheet numbers: the pin placed on the map's green from them. A position that cannot be the hole (a cup taken from the track; a mark that is not on the hole's own green) is not used, and the screen says so. Shot 1 from the map reads the scorecard yardage in the engine, as it already does on the row.

Is not: a change to the benchmark tables or the baseline (`js/analysis/benchmarks.js`, `tour-benchmark.js`), the category rules, the strokes-gained formula, the export format, `schemaVersion` or `migrate()`, any stored round, the course learning model (`learnTee`, `learnCup`, `learnGreen`, `rebuildCourseLearning`, `accumulatedHolePosition`), the live HUD, the recorder, the GPS pipeline, the shot-places preselection (v32), or the course map and its generator. A course with no map behaves exactly as it does at 7d35028.

1. What exists today `[measured]`, at 7d35028

Piece	Where	What it does
Where the hole is	holePosition(hole, fallback), js/round/round.js 447-469	Tier 1 hole.cup of ANY method, source cup. Tier 2 the ball marked on the green, source ball-on-green. Tier 3 fallback, which every caller fills with accumulatedHolePosition.
Per-shot distances	shotGeometry(hole, fallbackPos), round.js 471-498	A typed distance (distanceFt) wins; else the shot's mark to the hole position; else null. lengthM is the mark to the next mark, or to hole.cup.
The engine	js/analysis/strokes-gained.js: holeStates 78, holeStrokesGained 103, roundStrokesGained 228	Looks up E(lie, distance). A shot with no distance is unattributed. Takes fallbackFor(hole) from the caller.
The callers	js/analysis/trends.js 56-58 and 285-287; js/ui/screen-summary.js 197-199, 350; js/ui/screen-play.js 820	All pass accumulatedHolePosition(app, courseId, hole.number). None passes the map.
The map's green	toGreen(geometry, holeNumber, pos), js/round/course-geometry.js 129-168	Front, centre and back from a position. Used by the HUD and the end-of-hole rows, not by the engine.
The old paced pin	locateCupFromPaces, track-analysis.js 861-981; the "Where was the pin?" control, screen-play.js 1891-1977; setCupFromPaces, round.js 235	Paces on from the front edge and paces left or right of the MIDDLE, placed from where he walked on the track; needs a marked approach shot. Stored as hole.cup with method: 'paces' and cup.pinSheet.
The map tee (v32)	addMapTee, round.js 181	Shot 1 with source: 'map', mark.method: 'map' at the box centre, or no mark. The engine reads the position like any other mark: straight line from the box centre to the hole position.
Edit mode	screen-play.js 275	geometry is null in edit mode, so a finished round has no map for the end-of-hole entry or the rows.
The benchmark's distance	js/analysis/tour-benchmark.js, Table 9 note	"Distance to the hole in YARDS, measured along the fairway (dogleg distance from the tee), not straight line."
A new optional key	js/data/schema.js, device.recorder	Precedent: "Additive and optional: schemaVersion does not move and migrate() does not change". Export writes rounds verbatim; importExport saves migrate(round) and validates no hole key.

2. What the data says [measured]

2.1 The cups on his phone against the map

All 16 Veenker rounds in the phone's store as pulled 2026-09-28 (docs/roundDownloads/native-0927/localStorage-20260928.json).

Cups	n	On or within 5 yd of their own green	Distance to the map's green centre
Marked with a burst	35	31	of the 31: median 5.5 yd, largest 17.3 yd
Taken from the track (all 2026-09-27)	8	1	median 32.8 yd, largest 400.7 yd

The 4 burst cups that are not on their green: 7.7 yd outside (2026-09-11 hole 3), 93 yd outside (2026-09-27 hole 12, the accidental mark), 3,895 yd outside (2026-09-26 hole 1) and 3,983 yd outside (2026-09-16 hole 10). The engine uses every one of them today.

2.2 The greens on the map

18 holes. Every hole's line ends inside its own green, 0.1 to 4.1 yd from the centre. Depth along the line of play 17.5 to 42.4 yd (hole 11, hole 17). The farthest edge from the centre is 11.1 to 22.5 yd away. The last segment of the hole line is 11 to 280 yd long; on holes 3, 9, 17 and 10 it is 11, 18, 24 and 31 yd, too short to give a direction (hole 9: 12.7 degrees between the last segment and the line to the centre).

2.3 What a wrong distance costs, from the engine's own scratch table

Change in expected strokes for a 5 yd error in the distance to the hole:

Lie	15 yd	30 yd	50 yd	80 yd	120 yd	160 yd	200 yd
Fairway	0.114	0.052	0.026	0.013	0.014	0.022	0.031
Rough	0.130	0.049	0.034	0.016	0.017	0.021	0.028

From the tee, a 10 yd error is 0.010 to 0.067 strokes (150 to 500 yd).

2.4 What follows

- The green centre is within a median 5.5 yd of where the cup was (n = 31). That is a small error on an approach and a large one on a chip.
- On a hole whose tee reads the scorecard and whose first putt he typed, the strokes gained from tee to green ADD UP to the same total wherever the hole is taken from: the middle distances cancel in the sum. Only the split between off the tee, approach and short game moves. Section 7 test 9 proves it.
- 3 of 35 burst cups are nowhere near a green. A hole position has to be checked against the map before it is used.

3. Where the hole is [design]

3.1 The order, on a course with a map

The first of these that exists and passes 3.2:

Tier	Position	source	uncertaintyM
1	The cup he marked: <code>hole.cup</code> , any method except track	cup	<code>hole.cup.accuracyM</code> , as today
2	His pin sheet placed on the map's green (Section 4)	pin-sheet	4
3	The ball he marked on the green, as today	ball-on-green	the first putt plus the mark's accuracy, as today
4	The centre of the map's green	map-green	half the green's depth along the line of play (4.1)

The accumulated position from earlier rounds is not reached on a course with a map: tier 4 always exists.

3.2 A position that cannot be the hole is not used

- **A cup taken from the track** (`hole.cup.method === 'track'`) is never used on a course with a map. His item 3.
- **A cup or a ball mark more than 15 m from the hole's own green** (outside the green's ring by more than `HOLE_ON_GREEN_M = 15`) is not used. The next tier is.
- **A pin sheet that lands more than 3 m outside the green's ring**, or that cannot be placed (4.2), is not used. Tier 3 or 4 is.
- Nothing is deleted or rewritten. The mark stays on the hole, the pin sheet stays as he typed it. This is a filter at read time.
- Every position passed over is returned in skipped, one entry each: { `what: 'cup' | 'ball' | 'pin-sheet'`, `why: 'from-track' | 'off-green' | 'not-placed'`, `offM` } (`offM` null when it does not apply). The screens show it (Section 6).
- A hole the map has no green for is treated as a hole on a course with no map (3.3).

3.3 A course with no map

Exactly today's order and today's numbers: `hole.cup` of any method, the ball on the green, the accumulated position. No filter, no pin sheet.

3.4 The module

A new file, `js/round/hole-position.js`. It imports from `js/round/course-geometry.js`, `js/util/polygon.js` and `js/util/geo.js` and from nothing else; `js/round/course-geometry.js` is READ, not written (another session is adding to it).

Export	Returns
<code>HOLE_ON_GREEN_M = 15</code> , <code>PIN_OFF_GREEN_M = 3</code> , <code>PIN_SHEET_UNCERTAINTY_M = 4</code> , <code>APPROACH_BACK_M = 137.16</code>	constants

Export	Returns
greenFrame(geometry, holeNumber)	{ centre, front, depthM, bearingDeg, ring, greenId } or null
pinFromSheet(geometry, holeNumber, pinSheet)	{ placed: true, lat, lon, offGreenM, fromCentreM } or { placed: false, why } or null with no map or no entry
offOwnGreenM(geometry, holeNumber, pos)	metres outside the green's ring, 0 inside; null with no map
resolveHolePosition(hole, { geometry = null, accumulated = null })	{ lat, lon, source, uncertaintyM, skipped: [] }, or null

holePosition and shotGeometry in round.js take the same second argument, { geometry, accumulated }, and holePosition returns what resolveHolePosition returns. roundStrokesGained's option fallbackFor is renamed contextFor and returns that object per hole. A helper in round.js, holeContextFor(app, round), builds it: (hole) => ({ geometry: courseGeometry(getCourse(app, round.courseId)), accumulated: accumulatedHolePosition(app, round.courseId, hole.number) }). All five call sites in Section 1 use it. [verify] no import cycle.

4. The pin sheet on the map [design]

4.1 The green's frame

1. **The approach point A:** the point on the hole's line that is 150 yd (APPROACH_BACK_M) back from the line's end, measured along the line; the line's start when the line is shorter than that. Reason: Section 2.2, the last segment alone is too short on four holes.
2. **The centre C:** ringCentroid of the green's ring, the same centre toGreen uses.
3. **The line of play:** the bearing from A to C.
4. **The front edge F:** the first crossing of the ray A to C with the green's ring (rayRingIntersections). **The depth:** last crossing minus first.
5. uncertaintyM of tier 4 is depthM / 2.

4.2 Placing the pin

The entry is { onPaces, side, sidePaces, sideFrom, paceFeet } (Section 5). One pace is paceFeet x 0.3048 m.

1. The point on the line of play onPaces from F, toward C.
2. Through that point, the line across the green at right angles. Its crossings with the ring give the left edge and the right edge there. Right is to the right of a player facing from A to C.
3. sideFrom: 'edge': side: 'L' is sidePaces in from the left edge, side: 'R' is sidePaces in from the right edge, side: 'C' is halfway between the two edges.
4. sideFrom: 'centre': sidePaces from the line of play, right for 'R', left for 'L', on it for 'C'. Supported by the function; the sheet in Part B writes 'edge'.
5. placed: false when the point in step 1 is not inside the ring, or either edge in step 2 has no crossing. why says which.
6. offGreenM is how far the placed pin is outside the ring (0 inside); fromCentreM is its distance from C.

5. The data model [design]

One new optional key on a hole, additive, by the device.recorder precedent: schemaVersion does not move, migrate() does not change, a round logged before it simply has no key.

```
hole.pinSheet = null | {
  onPaces: integer >= 0,           // paces on from the front edge
  side:    'L' | 'R' | 'C',
  sidePaces: integer >= 0 | null,   // null when side is 'C'
  sideFrom: 'edge' | 'centre',
  paceFeet: number,                // his stride when he typed it
```

```

    enteredAt: ISO string
  }

```

- newHole in js/data/schema.js gets pinSheet: null and the comment. That is the only change to schema.js.
- What is stored is what he typed. No latitude or longitude is stored: the pin is placed from the map when it is read, so a corrected map moves the pin with the green.
- hole.cup keeps meaning what it means today: a position he marked, or the old paced construction. Nothing in this job writes hole.cup.
- A reader that finds hole.pinSheet undefined treats it as null.
- setPinSheet(hole, entry) and clearPinSheet(hole) in round.js. They do not touch shots, cup, greenEntry or completedAt. holeWindow does not read pinSheet.
- The export and the import are not changed: the key rides inside the round. [verify] a round with pinSheet survives buildExport then importExport with the key intact.

6. The numbers that depend on it [design]

6.1 Shot 1 from the map reads the scorecard in the engine

In shotGeometry, a shot with source === 'map' and lie === 'tee':

- toHoleM is hole.yards x 0.9144; toHoleSource is 'scorecard'; toHoleUncertaintyM is null. With no hole.yards, null.
- lengthM is null. A drive length needs his own mark at both ends; the box centre is not his mark. The "Tee shots (measured)" card therefore does not count it.

Reason: the row already reads "(scorecard)", and the benchmark's tee distance is the hole's length along the fairway, not a straight line (Section 1). A tee he marked himself is unchanged.

6.2 The end of the last shot

In shotGeometry, end for a shot with no next mark is the resolved position only when its source is cup; otherwise null. A cup that 3.2 passes over does not measure a shot's length.

6.3 What the engine returns

- Every shot in holeStrokesGained(...).shots already carries distanceSource; it now also carries distanceUncertaintyYd.
- roundStrokesGained adds sources: for the shots that are not putts, how many took their distance from each source (scorecard, cup, pin-sheet, ball-on-green, map-green, yards and the other typed units, accumulated-cup, accumulated-green), and unknown for a shot with no distance.
- roundStrokesGained adds positionNotes: the skipped entries of every hole scored, one per hole and reason.
- The formula, the categories, the putting identity and unattributed are not changed.

6.4 What the screens say

Screen	Change
Strokes gained card, screen-summary.js	Under the total, one line from sources, every count with its source: for example "Distance to the hole: 9 tees from the scorecard, 14 shots to the centre of the map's green, 3 to your pin sheet, 2 you typed." Then one line per reason in positionNotes, holes listed: for example "Holes 1, 10, 12: the marked cup is not on that hole's green on the map, so the map's green was used." Then, when any shot used map-green: "The centre of the green is a median 5.5 yd from where the cup was (n = 31 cups marked at Veenker). The total does not depend on it; the split between approach and short game does." The numbers in that last line are constants from Section 2.1 with a comment naming this spec.
Data quality card, screen-summary.js 441-462	"Hole position" counts holes by the resolved source, all of 3.1's sources. The sentence about the accumulated positions shows only on a course with no map.

Screen	Change
Shot list, screen-play.js 820-861	The second line of a shot reads scorecard for a map tee, pin sheet for pin-sheet, green centre, est. +/-n yd for map-green. The accumulated wording stays for a course with no map.
End-of-hole rows, shotRowHeading	The distance is to the resolved position: no suffix for cup, "(pin sheet)" for pin-sheet, "(ball on the green)" for ball-on-green, "(green centre)" for map-green. The number on the row is the number the engine uses.
Edit mode, screen-play.js 275	The map is available to the end-of-hole entry, the rows and the shot list in edit mode. The live HUD stays off in edit mode, as today.

7. Part A - tests that prove it

In the browser suite (/test/), one focused test per behaviour, sized like the neighbouring strokes-gained and geometry tests.

#	Behaviour	Test
1	No cup on a map course	a Veenker hole, track shots, no cup: every off-green shot has a distance, distanceSource is map-green, the distance equals toGreen(...).centreM
2	A marked cup wins	a burst cup 6 yd from the centre: source cup, distances to the cup
3	A cup from the track is not used	cup.method: 'track' inside the green: source map-green, one skipped entry
4	A cup off the green is not used	a burst cup 93 yd outside: source map-green, one skipped entry; the same cup 10 m outside: source cup
5	The pin sheet places the pin	hole 1, { onPaces: 12, side: 'L', sidePaces: 5, sideFrom: 'edge', paceFeet: 3 }: the pin is 12 yd from F along the line and 5 yd inside the left edge, within 0.2 m; source pin-sheet
6	side: 'C' and sideFrom: 'centre'	C is halfway between the edges; centre R 4 is 4 yd right of the line
7	A pin sheet that cannot be the hole	onPaces 5 more than the depth in paces: placed: false or off the green, source falls to map-green, one skipped entry
8	The map tee reads the scorecard	a source: 'map' tee on hole 15 (386 yd): toHoleM is 386 yd, source scorecard, lengthM null; a source: 'gps' tee is unchanged
9	The total does not move with the pin	one hole, map tee, two track shots, first putt typed: the sum of the three off-green strokes gained is equal to 1e-9 with no pin sheet, with a pin at the front and with a pin at the back; the approach and short game values differ
10	No map, no change	a Radcliffe hole with a track cup and an accumulated position: holePosition and the strokes gained are identical to the values the same round gives at 7d35028 (fixture values written into the test)
11	The key is optional and survives the export	a hole with no pinSheet key resolves like pinSheet: null; a round with one survives buildExport and importExport unchanged
12	The frame	hole 9's line of play is the bearing from the point 150 yd back along its line to the centre, not its 18 yd last segment; hole 11 (a 152 yd line) uses the line's start
13	The sources are counted	roundStrokesGained(...).sources adds up to the number of shots that are not putts; positionNotes carries test 4's hole

8. Part A - evidence in the report

- Before and after, every Veenker round in the phone's store**
(docs/roundDownloads/native-0927/localStorage-20260928.json, read only): holes scored, strokes unattributed, the four category totals and the total, at 7d35028 and at the new build, with the sources counts. One row per round that scores at least one hole. No round is written.
- The round trip, n = 31:** for each burst cup on or within 5 yd of its green (Section 2.1), the pin sheet numbers that describe it (whole paces, paceFeet 3, from the edge), placed back with pinFromSheet, and the distance from the placed pin to the cup: median and largest. This checks the arithmetic, not the course: it is not a field test of a real pin sheet.
- The frame, 18 holes:** bearing, depth and front-to-centre per hole.
- Suite counts. Time and tokens.

9. Part B - the PIN SHEET sheet [design]

Built after Fable has signed Part A.

1. **Where:** the Round menu (the three-line button) gets **Pin sheet**, in a live round and in edit mode. The round summary gets **PIN SHEET** under EDIT / ADD HOLES; it opens the round in edit mode with the sheet up. Only on a course with a map.
2. **The sheet:** one row per hole of the round, in the order played. Each row: the hole number; the green's depth from the map in paces ("25 deep"); ON, a number field; L / C / R; the side number, a number field, disabled on C. One line at the top: "Paces on from the front edge, then paces from the left or right edge. Your stride is set to n ft."
3. **The read-out** under a row once ON has a number: "n yd from the centre" when it places on the green; "lands n yd off the green on the map - check the numbers" or the why of placed: false, in the warning colour, when it does not. He can save either way: what he typed is what is stored.
4. **SAVE** writes hole.pinSheet for every row with an ON number (sideFrom: 'edge', paceFeet from his settings) and clears it on a row whose ON is empty. One UNDO, through the banner that stays (v31), puts every hole's pin sheet back as it was.
5. **Limits:** ON 0 to 60, side 0 to 30, whole numbers. Anything else is not accepted by the field.
6. **The old control:** on a course with a map the green sheet's "Where was the pin?" section is not shown; the PIN SHEET sheet is the one place. On a course with no map it is unchanged. Nothing else in the green sheet moves.
7. **Fit:** every row and SAVE reachable at 360x728, no sideways scroll, the number fields at least 44 px tall.

Tests, one each: the menu and the summary open the sheet on Veenker and not on Radcliffe; typing 12 / L / 5 on hole 1 and SAVE stores exactly Section 5's object; an empty ON clears the key; UNDO restores; the read-out warns on a pin off the green and SAVE still stores it; the green sheet has no "Where was the pin?" on Veenker and has it on Radcliffe; the fit at 360x728.

10. What Opus must not touch

android/, the recorder, the GPS pipeline (js/gps/gps.js, js/util/geo.js), js/util/polygon.js, js/round/course-geometry.js, js/data/geometry/, tools/course-geometry/, js/analysis/benchmarks.js, js/analysis/tour-benchmark.js, the category rules and the formula in strokes-gained.js, js/data/store.js, trackstore.js, persistence.js, schemaVersion and migrate(), the course learning functions, proposeHoleShots and its constants, the green sheet's putt entry, any stored round, docs/roundDownloads/, REVISION, the push. BUILD.id and the gt-shell-<id> cache move together.

11. Known, and not in this job

- The live HUD still reads the green's centre when a pin sheet is entered.
- firstPuttM and the green sheet's GPS first putt still read hole.cup of any method.
- rebuildCourseLearning learns a cup of any method, and a cup that is not on its green.
- A tee he marked himself reads a straight line to the hole, short of the card on a dogleg (hole 15: 19.5 percent, [measured] 2026-09-26).
- A drive from a map tee has no length. Whether an estimate from the tee box, labelled as one, is wanted is his call.
- The benchmark table's scratch-versus-Tour question (docs/benchmark-verification.md) is untouched.